TOR AMPLITUDE CAN BE CHECKED ON THE STALKS WHEN THE SITE HAS ENOUGH POINTS, AND ALWAYS PASSES TO STALKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full for both parts: the pullback interpretation of the stalk and the stalk criterion for the converse.

Editorial note. The label lemma-tor-amplitude-stalk names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Tor amplitude and its behaviour under pullback. Same chapter, the definition of tor amplitude in an interval and the lemma stating that it is preserved by pullback along a morphism of ringed sites. Supplies the reduction in part (1), where taking the stalk at a point is a pullback of ringed sites.
- Exactness can be checked on stalks. Modules on Sites, the lemma allowing the vanishing of the cohomology of a tensor product to be tested at the points of the site. Supplies the converse in part (2) when the site has enough points.

Source reference. Original reference 08H5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 12543–12543. Prerequisite pointer invoked at proof line(s) 12706; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05V3: sources/stacks-chow/tags.txt, source lines 7537–7537. Prerequisite pointer invoked at proof line(s) 12713; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let E be an object of $D(\mathcal{O})$. Let $a, b \in \mathbf{Z}$.*

(1) *If E has tor amplitude in $[a, b]$, then for every point p of the site $\mathcal{C}$ the object E_p of $D(\mathcal{O}_p)$ has tor amplitude in $[a, b]$.*
(2) *If $\mathcal{C}$ has enough points, then the converse is true.*

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Proof. Proof of (1). This follows because taking stalks at p is the same as pulling back by the morphism of ringed sites $(p, \mathcal{O}_p) \to (\mathcal{C}, \mathcal{O})$ and hence we can apply Lemma 08H5.

Proof of (2). If $\mathcal{C}$ has enough points, then we can check vanishing of $H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F})$ at stalks, see Modules on Sites, Lemma 05V3. Since $H^i(E \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F})_p = H^i(E_p \otimes^{\mathbf{L}}_{\mathcal{O}_p} \mathcal{F}_p)$ we conclude. □